



Chitkara

PAPER OF CSE-2025 - PYTHON TEST8 EXAM

Q1. Galactic Trading Hub Credit Adjustment

In the **Galactic Trading Hub of Nebula-7**, merchants list prices of various interplanetary goods in **Galactic Credits (GC)**.

Due to fluctuations in cosmic energy flow, the Interstellar Trade Council updates the pricing rules:

- Items priced **below 100 GC** → receive a **10% energy-boosted price increase**.
- Items priced **above 500 GC** → face a **5% gravity-tax deduction**.
- All other items (100–500 GC) remain stable under neutral cosmic conditions.
- After adjustments, all prices must be **rounded to the nearest whole credit**.

Your task is to apply these revisions and return the updated list of prices.

Input Format:

A list of integers → galactic item prices (in GC)

Output Format:

A list of updated prices (integers)

Sample Input 1

[75, 600, 150, 40, 520, 480]

Sample Output 1

[82, 570, 150, 44, 494, 480]

Sample Input 2

[45, 500, 501, 250, 95, 720]

Sample Output 2

[50, 500, 476, 250, 105, 684]

ANSWER:

Test case 1 :

Input:

[75, 600, 150, 40, 520, 480]

Output:

[82, 570, 150, 44, 494, 480]

Test case 2 :

Input:

[45, 500, 501, 250, 95, 720]

Output:

[50, 500, 476, 250, 105, 684]

Test case 3 :

Input:

[99, 101, 501, 500]

Output:

[109, 101, 476, 500]

Test case 4 :

Input:

[80, 90, 95, 99]

Output:

[88, 99, 105, 109]

Test case 5 :

Input:

[600, 650, 700]

Output:

[570, 618, 665]

Test case 6 :

Input:

[100, 200, 300, 400, 500]

Output:

[100, 200, 300, 400, 500]

Q2. Movie Theatre Ticket Price Revision

A multiplex revises its movie ticket prices across different seating categories due to maintenance and operational changes. The management introduces the following rules:

Ticket Price Update Rules

- Tickets **priced below ₹100** → increased by **10%**
- Tickets **priced above ₹500** → reduced by **5%**
- Tickets **priced between ₹100 and ₹500** → remain unchanged
- All updated ticket prices must be **rounded to the nearest integer**

Input Format

A list of integers representing ticket prices.

Output Format

A list of updated ticket prices (integers) after applying the rules.

Sample Input 1

[85, 450, 510, 95, 300, 700]

Sample Output 1

[94, 450, 484, 105, 300, 665]

Sample Input 2

[60, 120, 520, 100, 99, 800]

Sample Output 2

[66, 120, 494, 100, 109, 760]

ANSWER:

Test case 1 :

Input:

[85, 450, 510, 95, 300, 700]

Output:

[94, 450, 484, 105, 300, 665]

Test case 2 :

Input:

[60, 120, 520, 100, 99, 800]

Output:

[66, 120, 494, 100, 109, 760]

Test case 3 :

Input:

[99, 101, 501, 500]

Output:

[109, 101, 476, 500]

Test case 4 :

Input:

[80, 90, 95, 99]

Output:

[88, 99, 105, 109]

Test case 5 :

Input:

[600, 650, 700]

Output:

[570, 618, 665]

Test case 6 :

Input:

[100, 200, 300, 400, 500]

Output:

[100, 200, 300, 400, 500]

Q3. Electricity Board Tariff Adjustment

The city's Electricity Board is revising household electricity tariffs due to seasonal demand changes.

To ensure fair and balanced billing, tariffs are updated based on the following rules:

Tariff Update Rules

- Tariffs **below ₹100** → increased by **10%**
- Tariffs **above ₹500** → decreased by **5%**
- Tariffs between **₹100 and ₹500** (inclusive) → **remain unchanged**
- After revision, each tariff must be **rounded to the nearest integer**

Your Task

Write a program that updates the electricity tariffs according to the rules above and returns the final list of revised tariffs.

Input Format

A list of integers representing electricity tariffs.

Output Format

A list of updated tariffs (integers) after applying the revision rules.

Sample Input

[75, 120, 540, 90, 600]

Sample Output

[82, 120, 513, 99, 570]

Sample Input 2

[88, 150, 510, 40, 560, 100]

Sample Output 2

[97, 150, 484, 44, 532, 100]

ANSWER:

Test case 1 :

Input:

[75, 120, 540, 90, 600]

Output:

[82, 120, 513, 99, 570]

Test case 2 :

Input:

[88, 150, 510, 40, 560, 100]

Output:

[97, 150, 484, 44, 532, 100]

Test case 3 :

Input:

[99, 101, 501, 500]

Output:

[109, 101, 476, 500]

Test case 4 :

Input:

[80, 90, 95, 99]

Output:

[88, 99, 105, 109]

Test case 5 :

Input:

[600, 650, 700]

Output:

[570, 618, 665]

Test case 6 :

Input:

[100, 200, 300, 400, 500]

Output:

[100, 200, 300, 400, 500]

Q4. Game Score Submission Tracker

A gaming platform tracks high scores submitted by users. Each submission logs the **Player Handle**, the **Game Level**, and the **Score Achieved**. Sometimes, automated bots submit garbage data instead of an actual score.

You must:

1. Calculate the **total aggregate Score Achieved** for each Player Handle.
2. For any submission where the Score Achieved is **not a valid integer**, print:

Invalid record skipped

3. Print the final aggregated scores dictionary.

Input Format

A **Python list of tuples**, where each tuple is:

(Player Handle, Game Level, Score Achieved)

Output Format

Print:

- "Invalid record skipped" for each invalid score
- The final dictionary in the format:

```
{PlayerHandle : TotalScore}
```

Sample Input :

```
[('Max_P', 'L1', 100), ('Ray_G', 'L2', 50), ('Max_P', 'L3', 200), ('Ray_G', 'L4', 'score')]
```

Sample Output :

```
Invalid record skipped  
{'Max_P': 300, 'Ray_G': 50}
```

ANSWER:

Test case 1 :

Input:

```
[('Max_P', 'L1', 100), ('Ray_G', 'L2', 50), ('Max_P', 'L3', 200), ('Ray_G', 'L4', 'score')]
```

Output:

```
Invalid record skipped  
{'Max_P': 300, 'Ray_G': 50}
```

Test case 2 :

Input:

```
[('A','S','x')]
```

Output:

```
Invalid record skipped  
{}
```

Test case 3 :

Input:

```
[('A','S',1)]
```

Output:

```
{'A': 1}
```

Test case 4 :

Input:

```
[('A','S',5),('A','D',5)]
```

Output:
{'A': 10}

Test case 5 :

Input:
[('A','S',5),('B','S',5)]
Output:
{'A': 5, 'B': 5}

Test case 6 :

Input:
[('A','S',5),('A','D',15)]
Output:
{'A': 20}

Q5. Airline Baggage Weight Manifest

An airline creates a manifest logging luggage weights. Each entry notes the **Flight Number**, the **Passenger Class**, and the **Baggage Weight** (in kg). Sometimes, the scales produce unreadable text output.

You must:

- Calculate the **total Baggage Weight** carried on each **Flight Number**.
- **Only integer weights are valid; skip** any record with a non-integer weight.
- Output the message **"Corrupt manifest entry detected"** for every skipped record.

Input Format:

A list of tuples with values in form: (Flight Number, Passenger Class, Baggage Weight)

Output Format:

Dictionary format output {Flight Number : Total Weight (kg)}

Sample Input 1:

```
[('F300', 'Eco', 20), ('F400', 'Bus', 30), ('F300', 'Bus', 15)]
```

Sample Output 1:

```
{'F300': 35, 'F400': 30}
```

Sample Input 2:

```
[('F300', 'Eco', 20), ('F400', 'Bus', 30), ('F300', 'Bus', 15), ('F500', 'Eco', 'fifty')]
```

Sample Output 2:

```
Corrupt manifest entry detected  
{'F300': 35, 'F400': 30}
```

ANSWER:

Test case 1 :

Input:

```
[('F300', 'Eco', 20), ('F400', 'Bus', 30), ('F300', 'Bus', 15), ('F500', 'Eco', 'fifty')]
```

Output:

```
Corrupt manifest entry detected  
{'F300': 35, 'F400': 30}
```

Test case 2 :

Input:

```
[('A','S','x')]
```

Output:

```
Corrupt manifest entry detected  
{}
```

Test case 3 :

Input:

[('A','S',1)]

Output:

{'A': 1}

Test case 4 :

Input:

[('A','S',5),('A','D',5)]

Output:

{'A': 10}

Test case 5 :

Input:

[('A','S',5),('B','S',5)]

Output:

{'A': 5, 'B': 5}

Test case 6 :

Input:

[('A','S',5),('A','D',15)]

Output:

{'A': 20}

Q6. Marketing Campaign Clicks

A marketing team tracks the performance of various ads. Each log entry records the **Ad ID**, the **Platform Used**, and the number of **Clicks** received in a session. Due to platform logging errors, the Clicks count is sometimes recorded as text.

You must:

- Determine the **Total Clicks** received for each **Ad ID**.
- Ignore all records where the Clicks value is non-numeric and print **Invalid record skipped**
- Print the output as a dictionary.

Input Format:

A list of tuples: (Ad ID, Platform Used, Clicks)

Output Format:

Dictionary {Ad ID : Total Clicks}

Sample Input 1:

[('A1', 'FB', 500), ('A2', 'Insta', 100), ('A1', 'Google', 250)]

Sample Output 1:

{ 'A1': 750, 'A2': 100 }

Sample Input 2:

```
[('A1', 'FB', 500), ('A2', 'Insta', 100), ('A1', 'Google', 250), ('A3', 'FB', 'one hundred')]
```

Sample Output 2:

```
Invalid record skipped  
{'A1': 750, 'A2': 100}
```

ANSWER:Test case 1 :

Input:

```
[('A1', 'FB', 500), ('A2', 'Insta', 100), ('A1', 'Google', 250), ('A3', 'FB', 'one hundred')]
```

Output:

Invalid record skipped

```
{'A1': 750, 'A2': 100}
```

Test case 2 :

Input:

```
[('A','S','x')]
```

Output:

Invalid record skipped

```
{}
```

Test case 3 :

Input:

```
[('A','S',1)]
```

Output:

```
{'A': 1}
```

Test case 4 :

Input:

```
[('A','S',5),('A','D',5)]
```

Output:

```
{'A': 10}
```

Test case 5 :

Input:

```
[('A','S',5),('B','S',5)]
```

Output:

```
{'A': 5, 'B': 5}
```

Test case 6 :

Input:

```
[('A','S',5),('A','D',15)]
```

Output:

```
{'A': 20}
```

Q7. Roller Coaster Design Drop Chain

A theme park designer is reviewing a roller coaster track plan.

The track is represented as a string where:

- 'U' = Uphill or flat segment
- 'D' = Drop segment

Your task is to find the **length of the longest continuous chain of 'D' segments**.

Input Format

A single string `track_sequence` consisting of characters 'U' and 'D'.

Output Format

Print the maximum length of the longest continuous sequence of 'D' characters.

Constraints

$1 \leq \text{len}(\text{track_sequence}) \leq 10^5$
`track_sequence` contains only 'U' and 'D'

Sample Input:

```
"UUDUDDDDUDDUUUDDDD"
```

Sample Output:

```
4
```

ANSWER:

Test case 1 :

Input:

```
UUDUDDDDUDDUUUDDDD
```

Output:

```
4
```

Test case 2 :

Input:

```
UUUUUU
```

Output:

```
0
```

Test case 3 :

Input:

```
DDDDD
```

Output:

5

Test case 4 :

Input:

DUDUDUD

Output:

1

Test case 5 :

Input:

Output:

0

Test case 6 :

Input:

DD

Output:

2

Q8. Mining Ore Deposit Measurement

A mining geologist scans a rock face, marking points as either '**O**' (Ore) or '**R**' (Rock). They need to identify the richest, longest continuous ore deposit section.

Process the scan string and find the total length of the **maximum continuous deposit of 'O'** (Ore).

Input Format:

A string 'scan_string' representing the rock face scan.

Output Format:

Print the maximum continuous length of the Ore deposit.

Sample Input:

"RORROOOOROOOO"

Sample Output:

4

ANSWER:

Test case 1 :

Input:

RORROOOOROOOO

Output:

4

Test case 2 :

Input:

RRRRRR

Output:

0

Test case 3 :

Input:

OOOOO

Output:

5

Test case 4 :

Input:

RORORO

Output:

1

Test case 5 :

Input:

Output:

0

Test case 6 :

Input:

O

Output:

1

Q9. Chemical Reaction Stability Log

A chemist tracks the stability of a polymerization reaction, logging the state every few seconds as either '**S**' (**Stable**) or '**U**' (**Unstable**). They need to find the maximum time the reaction remained perfectly stable without interruption.

Find and print the length of the **longest continuous sequence of 'S'** (Stable) entries in the log string.

Input Format:

A string 'log_string' containing the stability log

Output Format:

Print the maximum continuous duration of power surges (consecutive 'S' characters) from the input

data.

Sample Input:

"USSUUSSSUUU"

Sample Output:

3

ANSWER:Test case 1 :

Input:

USSUUSSSUUU

Output:

3

Test case 2 :

Input:

UUUUUU

Output:

0

Test case 3 :

Input:

SSS

Output:

3

Test case 4 :

Input:

SUSUS

Output:

1

Test case 5 :

Input:

Output:

0

Test case 6 :

Input:

S

Output:

1